1.

Let $X_1 \ldots X_n$ be iid of X:

$$M_n = \max(X_1, \ldots X_n)$$

since k is largest value

$$P(M_n < k) = P(X_1, \ldots X_n)$$

$$P(M_n < k) = P(X < k)^n$$

since $P(X = k) = p(k) > 0$

$$P(X < k) = 1 - p(k)$$

Therefore

$$P(M_n < k) = (1 - p(k))^n$$

Take limits

$$\lim_{n \to \infty} (1 - p(k))^n = 0$$

$$P(M_n = k) = (1 - p(k))^n \to 1$$

Therefore

$$M_n \xrightarrow{P} k.$$

Suppose we choose arbitary sequences $c_n > 0$ and d_n

define $Z_n = \dfrac{M_n - d_n}{c_n}$

and

$$a_n = \frac{k - d_n}{c_n}$$

Since $M_n = k \to 1$

$$P(Z_n = a_n) \to 1$$

Suppose, for contradition, that Z_n converges in distribution to a non-degenerate rv Z.

Convergence in distribution implies tightness. Since almost all prob mass

is concentrated at a_n, the sequence (a_n) must bound.

Therefor, there exists a subsequence:

$$a_{n_j} \longrightarrow a$$

along this subsequence

$$Z_{n_j} \xrightarrow{P} a.$$

Hence limitty CDF must

$$H(x) = \begin{cases} 0, & x < a \\ 1, & x \geq a \end{cases}$$

Therefore, No non-degenerate limitty CDF exists.

2.

(a)

$$F_u(x) = P(X - u \leq x \mid X > u).$$

use conditional probability

$$F_u(x) = \frac{F(u+x) - F(u)}{1 - F(u)}$$

since $X \sim U(0,1)$,

$$F(x) = x, \quad 0 \leq x \leq 1$$

Therefore

$$F_u(x) = \frac{(u+x) - u}{1-u}$$

$$= \frac{x}{1-u}$$

GPD distribution

$$G_{\xi,\beta}(x) = 1 - \left(1 + \frac{\xi x}{\beta}\right)^{-1/\xi}$$

set $\xi = -1$:

$$G_{-1,\beta}(x) = 1 - \left(1 - \frac{x}{\beta}\right)^{-1}$$

Therefore, $G_{-1,\beta}(x) = \frac{x}{\beta}$

(compare $F_u(x) = \frac{x}{1-u}$ with $G_{-1,\beta}(x) = \frac{x}{\beta}$

choose $\beta(u) = 1-u$

then $F_u(x) = G_{-1,\beta(u)}(x)$

Therefore

$$\sup_{0 \le x \le 1-u} |F_u(x) - G_{-1,\beta(u)}(x)| = 0$$

hence $$\lim_{u \uparrow 1} \sup_{0 \le x \le 1-u} |F_u(x) - G_{-1,\beta(u)}(x)| = 0$$

(b)

Normal Excess CDF

Standard Normal CDF:

$$\Phi(x) = P(X \le x)$$

Let

$$\bar{\Phi}(x) = 1 - \Phi(x)$$

$$F_u(x) = \frac{\bar{\Phi}(u+x) - \bar{\Phi}(u)}{1 - \bar{\Phi}(u)}$$

$$= 1 - \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)}$$

For $\xi = 0$, GPD becomes the exp distribution

$$G_{0,\beta}(x) = 1 - e^{-x/\beta}$$

since

$$\beta(u) = \frac{1}{u}$$

we have

$$G_{0,1/u}(x) = 1 - e^{-ux}$$

therefore

$$|F_u(x) - G_{0,1/u}(x)| = \left| 1 - \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} - (1 - e^{-ux}) \right|$$

$$= \left| e^{-ux} - \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \right|.$$

Show $\sup_{x \geq \delta} \left| e^{-ux} - \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \right| \to 0$

Apply Normal Tail Bounds

provides:

$$\frac{z}{1+z^2} e^{-z^2/2} \leq \int_z^\infty e^{-y^2/2}\, dy \leq \frac{1}{z} e^{-z^2/2}$$

for $z > 0$

Since

$$\bar{\Phi}(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-y^2/2}\, dy$$

upper bound

$$\bar{\Phi}(u+x) \leq \frac{1}{\sqrt{2\pi}(u+x)} e^{-(u+x)^2/2}$$

lower bound

$$\bar{\Phi}(u) \geq \frac{1}{\sqrt{2\pi}} \frac{u}{1+u^2} e^{-u^2/2}$$

$$\frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \leq \frac{1+u^2}{u(u+x)} e^{-[(u+x)^2 - u^2]/2}$$

expand

$$(u+x)^2 - u^2 = 2ux + x^2$$

Hence,

$$\frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \leq \frac{1+u^2}{u(u+x)} e^{-ux - x^2/2}$$

since $x \geq \delta$,

$$u + x \geq u + \delta$$

and

$$e^{-ux - x^2/2} \leq e^{-u\delta - \delta^2/2}$$

Therefore

$$0 \leq \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \leq \frac{1+u^2}{u(u+\delta)} e^{-u\delta - \delta^2/2}$$

As $u \to \infty$

$$\frac{1+u^2}{u(u+\delta)} \to 1$$

and

$$e^{-u\delta - \delta^2/2} \to 0$$

Thus $\sup_{x \geq \delta} \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \to 0$

Apply Triangle Inequality

$$|F_u(x) - G_{0,1/u}(x)| = \left| e^{-ux} - \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \right|.$$

By Tri Ineq

$$\left| e^{-ux} - \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)} \right| \leq e^{-ux} + \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)}.$$

Therefore

$$\sup_{x \geq \delta} |F_u(x) - G_{0,1/u}(x)| \leq e^{-u\delta} + \frac{1+a^2}{u(u+\delta)} e^{-u\delta - \delta^2/2}$$

Both term $\to 0$ when $u \to \infty$

So, $\lim_{u \to \infty} \sup_{x \geq \delta} |F_u(x) - G_{0,1/u}(x)| = 0$